

MIDLANDS STATE UNIVERSITY



FACULTY OF BUSINESS SCIENCES
DEPARTMENT OF INFORMATION AND MARKETING SCIENCES

NAME : LYN NYANDORO

REG NUMBER : R266444Y

PROGRAM : MASTER OF COMMERCE DEGREE IN INFORMATION
SYSTEMS MANAGEMENT (BS30)

COURSE CODE : MIM736 – SOFTWARE ENGINEERING

LECTURER : DR. ZHOU

LEVEL : 1.2

QUESTION : PRACTICAL ASSIGNMENT: Design and implement a DevOps-
Enabled Qualification Verification System Using Git and CI/CD
Practices

ROLE AND CONTRIBUTIONS

I acted as full-stack engineer and DevOps lead for this practical. My work covers the Spring Boot API (domain model, JWT security, qualification lifecycle, hash-based verification, audit persistence), the Create React App interface, Docker Compose packaging, GitHub Actions quality gates, and the technical documentation.

CONCRETE DELIVERABLES:

- Domain model for users, qualifications and verification records, including validation rules (dates, NQF band, unique verification codes).
- SHA-256 canonical hashing so authenticity checks detect payload tampering.
- Role-based access: issuers register and revoke; verifiers search and verify; students view their own credentials; administrators have full access.
- Login-first SPA routing ('/' and '/login'), seeded campus catalog plus audit events, and agentic insights (rule-based, optionally Groq LLM-assisted).
- Automated tests ('*CredentialHashServiceTest*', '*QualificationServiceTest*', '*JwtServiceTest*', '*QualificationVerificationIT*') and Jest UI tests.
- CI pipeline that fails the build on Checkstyle violations or JaCoCo coverage below 50%, then builds Docker images.
- Technical report, Git workflow description, issue backlog and this contribution statement.

Git evidence is the commit graph on '*main*'/'*develop*'/'*feature/**'. After the project is pushed to GitHub, attach screenshots of Actions runs, pull requests and Issues to the printed submission pack.

LESSONS LEARNT

1. Pipelines make quality non-negotiable. Encoding Checkstyle and coverage as Maven '*verify*' goals means a green build is evidence, not a claim.
2. Canonical hashing is simpler than a full blockchain for a coursework system, and still gives a defensible integrity story that can be extended later.
3. Role design leaks into the UI. Hiding the register screen from verifiers avoids 403 errors and matches the security model.
4. CRA on modern Node needs '*--legacy-peer-deps*'. Pinning Node 20 in Docker and CI avoided the Node 23 peer-dependency friction seen on the development machine.

CHALLENGES

- Balancing academic report length (3,000 words) with a working product in one iteration.
- Achieving a JaCoCo gate without writing brittle tests that only exist to inflate coverage.
- Demonstrating pull requests and issue tracking before a GitHub remote is configured; the local branch history and *'docs/ISSUES.md'* are the offline equivalent, to be mirrored on GitHub for marking.
- Merge-conflict practice required a deliberate overlapping edit; the resolution is recorded in *'docs/CONFLICT_LOG.md'*.

GIT ACTIVITY

```
(base) lynnyandoro@Lyns-Laptop Qualification-Verification-System % git shortlog -sn
 22 Lyn Nyandoro
  7 tatlegend
(base) lynnyandoro@Lyns-Laptop Qualification-Verification-System % git log --author="Lyn" --stat
commit 6c396e25d3c16da87c53e7148b08ed9bf0011ec0
Author: Lyn Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 21:33:01 2026 +0200
```

Add campus demo data, mint UI, and distinct credential IDs.

Seed three universities with student accounts, programme-based credential IDs plus QVS verification codes, employer verify, and an admin directory.

Co-authored-by: Cursor <cursoragent@cursor.com>

```
README.md | 27 +-
.../main/java/com/mim736/qvs/config/DataSeeder.java | 270 ++++++++
.../java/com/mim736/qvs/config/H2SchemaAligner.java | 62 +++
.../java/com/mim736/qvs/config/SecurityConfig.java | 1 +
.../main/java/com/mim736/qvs/domain/Institution.java | 47 ++
.../java/com/mim736/qvs/domain/Qualification.java | 15 +-
backend/src/main/java/com/mim736/qvs/domain/Role.java | 3 +-
.../main/java/com/mim736/qvs/domain/StudentStage.java | 7 +
.../main/java/com/mim736/qvs/domain/UserAccount.java | 29 +-
.../com/mim736/qvs/repo/InstitutionRepository.java | 11 +
.../com/mim736/qvs/repo/QualificationRepository.java | 20 +-
.../com/mim736/qvs/repo/UserAccountRepository.java | 14 +
.../com/mim736/qvs/service/AdminDirectoryService.java | 54 ++
.../main/java/com/mim736/qvs/service/AuthService.java | 6 +-
.../java/com/mim736/qvs/service/CredentialIds.java | 110 ++++
.../com/mim736/qvs/service/QualificationService.java | 93 ++++
.../java/com/mim736/qvs/service/StudentService.java | 138 ++++++
.../com/mim736/qvs/service/VerificationService.java | 4 +
.../main/java/com/mim736/qvs/web/AdminController.java | 33 ++
.../com/mim736/qvs/web/InstitutionController.java | 25 +
.../com/mim736/qvs/web/QualificationController.java | 10 +-
.../java/com/mim736/qvs/web/StudentController.java | 52 ++
.../java/com/mim736/qvs/web/dto/AuthResponse.java | 30 +-
.../com/mim736/qvs/web/dto/EnrollStudentRequest.java | 78 +++
.../com/mim736/qvs/web/dto/InstitutionResponse.java | 30 ++
.../qvs/web/dto/InstitutionSummaryResponse.java | 38 ++
.../com/mim736/qvs/web/dto/QualificationRequest.java | 10 +
.../com/mim736/qvs/web/dto/QualificationResponse.java | 6 +
.../com/mim736/qvs/web/dto/StudentDetailResponse.java | 22 +
.../java/com/mim736/qvs/web/dto/StudentResponse.java | 63 +++
.../com/mim736/qvs/web/dto/UserSummaryResponse.java | 64 +++
.../com/mim736/qvs/config/H2SchemaAlignerTest.java | 31 ++
.../mim736/qvs/service/QualificationServiceTest.java | 6 +
.../mim736/qvs/web/QualificationVerificationIT.java | 95 ++++
docs/DEMO_SCRIPT.md | 6 +-

```

```

frontend/public/index.html | 16 +-
frontend/src/App.js | 11 +-
frontend/src/App.test.js | 9 +-
frontend/src/VerifyPage.test.js | 2 +-
frontend/src/components/CopyId.js | 17 +
frontend/src/components/Layout.js | 133 ++++
frontend/src/components/PersonCell.js | 23 +
frontend/src/index.css | 578 ++++++-----
frontend/src/index.js | 9 +-
frontend/src/pages/AdminPage.js | 158 +++++
frontend/src/pages/AuditPage.js | 19 +-
frontend/src/pages/DashboardPage.js | 93 +++-
frontend/src/pages/LoginPage.js | 9 +-
frontend/src/pages/RegisterPage.js | 5 +-
frontend/src/pages/RegisterQualificationPage.js | 48 +-
frontend/src/pages/SearchPage.js | 136 +++-
frontend/src/pages/StudentDetailPage.js | 73 +++
frontend/src/pages/StudentsPage.js | 171 ++++++
frontend/src/pages/VerifyPage.js | 21 +-
frontend/src/theme.js | 60 +++
55 files changed, 2797 insertions(+), 304 deletions(-)

```

commit a90e4cfb1cd1504d9413f529c2922b0baee0fad4

Merge: a64bcd6 34c5ac8

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:53:18 2026 +0200

Promote UI hardening onto main.

Include the CRA ajv pin so markers can run npm start without a module error.

commit 34c5ac8091d3340b9fe355cbfb20d26bd30466a5

Merge: ab25c09 cb34dfb

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:53:18 2026 +0200

Merge feature/ui-hardening into develop.

Keep the UI start path working on current Node while preserving the session.

commit cb34dfbcd4130fd3c93a2c37abaec908efaa5cf1

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:53:18 2026 +0200

Harden the CRA login flow and pin ajv for webpack.

Redirect authenticated users away from the login page and keep the dev server compatible with modern npm hoisting.

Co-authored-by: Cursor <cursoragent@cursor.com>

```

frontend/package-lock.json | 149 ++++++-----
frontend/package.json | 1 +
frontend/src/pages/LoginPage.js | 8 +-
3 files changed, 128 insertions(+), 30 deletions(-)

```

commit a64bcd671750beacb4d61ceab6df43ecc8ee6ff8

Merge: 8e8e77c ab25c09

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:09 2026 +0200

Release develop into main for the MIM736 practical.

Promote the tested qualification verification system as the releasable line.

commit ab25c09ab3f95133e28583c11924dcffcd2c461

Merge: 571ac55 1857a3f

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:09 2026 +0200

Resolve overlapping documentation edits on the conflict log.

Keep both the hashing and audit-trail sentences so neither requirement is lost.

Co-authored-by: Cursor <cursoragent@cursor.com>

commit 571ac559ed2cfd4abbd96140d3ee506eb11117d7

Merge: 07369a1 2f153c3
Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 19:51:09 2026 +0200

Merge hashing wording into develop.

commit 1857a3fee61ce2dc1036d6ace6b27d64c05ab012
Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 19:51:09 2026 +0200

Clarify audit-trail wording in the conflict-practice note.

Emphasise that verification history is append-only.

Co-authored-by: Cursor <cursoragent@cursor.com>

docs/CONFLICT_LOG.md | 4 ++--
1 file changed, 2 insertions(+), 2 deletions(-)

commit 07369a1c513601ff4920408b5f5a092c522d7926
Merge: 524daeb 3c712af
Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 19:51:09 2026 +0200

Merge feature/documentation into develop.

Attach written evidence to the same repository as the running system.

commit 2f153c335ba88ad07e2cff5909965e8824486aba
Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 19:51:09 2026 +0200

Clarify hashing wording in the conflict-practice note.

Keep the integrity-hash explanation as the primary sentence for markers.

Co-authored-by: Cursor <cursoragent@cursor.com>

docs/CONFLICT_LOG.md | 4 ++--
1 file changed, 2 insertions(+), 2 deletions(-)

commit 524daeb55e3c0384cde9be301dd1bd8620e0e9e8
Merge: 6bcc28c bdbcl62
Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 19:51:09 2026 +0200

Merge feature/cicd-docker into develop.

Make integration, coverage and image build part of every change.

commit 3c712af961ee7bc2ef45f92bb38ca626103683d5
Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 19:51:09 2026 +0200

Document architecture, Git workflow, issues and assessment artefacts.

Capture the technical report, contribution statement and demonstration script for marking.

Co-authored-by: Cursor <cursoragent@cursor.com>

docs/CONFLICT_LOG.md | 7 ++
docs/DEMO_SCRIPT.md | 20 +++++
docs/GIT_WORKFLOW.md | 33 +++++++
docs/INDIVIDUAL_CONTRIBUTION.md | 51 ++++++++
docs/ISSUES.md | 22 +++++
docs/SUBMISSION.md | 26 ++++++
docs/TECHNICAL_REPORT.md | 191 +++++
7 files changed, 350 insertions(+)

commit 6bcc28cb8dada1852c66dffe0b494d6b32054c7d
Merge: 639d6d5 17eacd1
Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 19:51:08 2026 +0200

Merge feature/react-ui into develop.

Deliver the browser client required for the working-software demonstration.

commit bdbc162a92bc7f5506dfa656ad6faa57ed8b1976

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:08 2026 +0200

Add GitHub Actions quality gates and Docker Compose delivery.

Fail the pipeline on Checkstyle, test or coverage regressions and package a runnable stack.

Co-authored-by: Cursor <cursoragent@cursor.com>

```
.github/pull_request_template.md | 17 ++++++++
.github/workflows/ci.yml          | 66 +++++
backend/Dockerfile                | 13 +++++
docker-compose.yml               | 40 +++++
4 files changed, 136 insertions(+)
```

commit 639d6d548e5aa98733bc43ed8fc3d941096deaf2

Merge: 9adeb8b 785da56

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:08 2026 +0200

Merge feature/verification-audit into develop.

Complete the four functional requirements of the MIM736 brief on the API.

commit 17eacd174f26acd0cc3cbbb823c7eaa24164fc7a

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:08 2026 +0200

Add a Create React App interface for register, search, verify and audit.

Give authorised users a usable path through the same API the tests already cover.

Co-authored-by: Cursor <cursoragent@cursor.com>

```
frontend/.env.example           | 1 +
frontend/.nvrc                  | 1 +
frontend/Dockerfile             | 14 +
frontend/nginx.conf             | 23 +
frontend/package-lock.json      | 17807 +++++
frontend/package.json           | 46 +
frontend/public/index.html      | 13 +
frontend/src/App.js             | 58 +
frontend/src/App.test.js        | 16 +
frontend/src/VerifyPage.test.js | 30 +
frontend/src/api.js             | 18 +
frontend/src/auth.js            | 7 +
frontend/src/components/Layout.js | 42 +
frontend/src/index.css          | 259 +
frontend/src/index.js           | 14 +
frontend/src/pages/AuditPage.js | 49 +
frontend/src/pages/DashboardPage.js | 48 +
frontend/src/pages/LoginPage.js | 51 +
frontend/src/pages/RegisterPage.js | 72 +
frontend/src/pages/RegisterQualificationPage.js | 115 +
frontend/src/pages/SearchPage.js | 82 +
frontend/src/pages/VerifyPage.js | 67 +
frontend/src/setupTests.js      | 1 +
23 files changed, 18834 insertions(+)
```

commit 9adeb8b5be4ff198819a22d5e37bcf6846fe5d56

Merge: 96808b9 d855d10

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:08 2026 +0200

Merge feature/qualification-api into develop.

Expose registration and search as the core issuer and verifier workflows.

commit 785da56d5c618f287026b4c31c2a5fbeb14a228b3

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:08 2026 +0200

Verify authenticity and persist an append-only audit trail.

Record every check, including failures, so employers can evidence due diligence.

Co-authored-by: Cursor <cursoragent@cursor.com>

```
.../com/mim736/qvs/domain/VerificationRecord.java | 123 ++++++
.../com/mim736/qvs/domain/VerificationResult.java | 10 ++
.../mim736/qvs/repo/VerificationRecordRepository.java | 11 ++
.../com/mim736/qvs/service/VerificationService.java | 138 ++++++
.../com/mim736/qvs/web/VerificationController.java | 44 ++++++
.../com/mim736/qvs/web/dto/AuditRecordResponse.java | 65 ++++++
.../java/com/mim736/qvs/web/dto/VerifyRequest.java | 32 ++++
.../java/com/mim736/qvs/web/dto/VerifyResponse.java | 35 ++++++
backend/src/main/resources/application-docker.yml | 18 +++
.../mim736/qvs/web/QualificationVerificationIT.java | 181 ++++++
backend/src/test/resources/application.yml | 18 +++
11 files changed, 675 insertions(+)
```

commit 96808b9df65e7b60d354a45ee5813b3903d622ea

Merge: 8e8e77c f080609

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:08 2026 +0200

Merge feature/auth-and-roles into develop.

Integrate the security foundation so later qualification endpoints can reuse JWT roles.

commit d855d1007cf33ff2e02a121c53cac170593e5bb3

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:08 2026 +0200

Register and search qualifications with integrity hashing.

Store a canonical SHA-256 digest at registration so later authenticity checks can detect tampering.

Co-authored-by: Cursor <cursoragent@cursor.com>

```
.../main/java/com/mim736/qvs/config/DataSeeder.java | 74 ++++++
.../java/com/mim736/qvs/domain/Qualification.java | 199 ++++++
.../com/mim736/qvs/domain/QualificationStatus.java | 7 +
.../java/com/mim736/qvs/domain/QualificationType.java | 8 +
.../com/mim736/qvs/repo/QualificationRepository.java | 37 ++++
.../com/mim736/qvs/service/CredentialHashService.java | 45 ++++++
.../com/mim736/qvs/service/QualificationService.java | 145 ++++++
.../com/mim736/qvs/web/QualificationController.java | 65 ++++++
.../com/mim736/qvs/web/dto/QualificationRequest.java | 95 ++++++
.../com/mim736/qvs/web/dto/QualificationResponse.java | 107 ++++++
.../mim736/qvs/service/CredentialHashServiceTest.java | 48 ++++++
.../mim736/qvs/service/QualificationServiceTest.java | 87 ++++++
12 files changed, 917 insertions(+)
```

commit f080609bacc7287a11fd3e4f0b2be21a407770a6

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:08 2026 +0200

Add JWT authentication and role-based access control.

Issuers, verifiers and administrators need distinct privileges before any qualification data is exposed.

Co-authored-by: Cursor <cursoragent@cursor.com>

```
backend/checkstyle.xml | 34 +++++
backend/pom.xml | 167 ++++++
.../src/main/java/com/mim736/qvs/QvsApplication.java | 12 ++
.../java/com/mim736/qvs/config/SecurityConfig.java | 75 ++++++
backend/src/main/java/com/mim736/qvs/domain/Role.java | 7 +
.../main/java/com/mim736/qvs/domain/UserAccount.java | 107 ++++++
.../com/mim736/qvs/repo/UserAccountRepository.java | 15 ++
.../mim736/qvs/security/JwtAuthenticationFilter.java | 45 ++++++
.../main/java/com/mim736/qvs/security/JwtService.java | 66 ++++++
.../mim736/qvs/security/QvsUserDetailsService.java | 37 ++++++
.../main/java/com/mim736/qvs/service/AuthService.java | 69 ++++++
.../main/java/com/mim736/qvs/web/AuthController.java | 35 ++++++
.../java/com/mim736/qvs/web/BusinessException.java | 12 ++
.../com/mim736/qvs/web/GlobalExceptionHandler.java | 44 ++++++
.../java/com/mim736/qvs/web/NotFoundException.java | 12 ++
.../java/com/mim736/qvs/web/dto/AuthResponse.java | 34 +++++
```

```

.../java/com/mim736/qvs/web/dto/LoginRequest.java | 28 +++
.../java/com/mim736/qvs/web/dto/RegisterRequest.java | 66 ++++++++
backend/src/main/resources/application.yml | 45 ++++++
.../java/com/mim736/qvs/security/JwtServiceTest.java | 25 +++
20 files changed, 935 insertions(+)

```

commit 8e8e77cd9eafb828cc64b18ed3d84cf19242bc2b

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:07 2026 +0200

Initialise the qualification verification repository.

Establish the assignment README and ignore rules so Java, React and Docker artefacts stay out of version control.

Co-authored-by: Cursor <cursoragent@cursor.com>

```

.gitignore | 43 ++++++
README.md | 78 ++++++
2 files changed, 121 insertions(+)
(END)
.../src/main/java/com/mim736/qvs/QvsApplication.java | 12 ++
.../java/com/mim736/qvs/config/SecurityConfig.java | 75 ++++++++
backend/src/main/java/com/mim736/qvs/domain/Role.java | 7 +
.../main/java/com/mim736/qvs/domain/UserAccount.java | 107 ++++++++
.../com/mim736/qvs/repo/UserAccountRepository.java | 15 ++
.../mim736/qvs/security/JwtAuthenticationFilter.java | 45 ++++++
.../main/java/com/mim736/qvs/security/JwtService.java | 66 ++++++++
.../mim736/qvs/security/QvsUserDetailsService.java | 37 ++++++
.../main/java/com/mim736/qvs/service/AuthService.java | 69 ++++++++
.../main/java/com/mim736/qvs/web/AuthController.java | 35 ++++++
.../java/com/mim736/qvs/web/BusinessException.java | 12 ++
.../com/mim736/qvs/web/GlobalExceptionHandler.java | 44 ++++++
.../java/com/mim736/qvs/web/NotFoundException.java | 12 ++
.../java/com/mim736/qvs/web/dto/AuthResponse.java | 34 ++++++
.../java/com/mim736/qvs/web/dto/LoginRequest.java | 28 ++++
.../java/com/mim736/qvs/web/dto/RegisterRequest.java | 66 ++++++++
backend/src/main/resources/application.yml | 45 ++++++
.../java/com/mim736/qvs/security/JwtServiceTest.java | 25 +++
20 files changed, 935 insertions(+)

```

commit 8e8e77cd9eafb828cc64b18ed3d84cf19242bc2b

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>

Date: Thu Sep 3 19:51:07 2026 +0200

Initialise the qualification verification repository.

Establish the assignment README and ignore rules so Java, React and Docker artefacts stay out of version control.

Co-authored-by: Cursor <cursoragent@cursor.com>

```

.gitignore | 43 ++++++
README.md | 78 ++++++
2 files changed, 121 insertions(+)
(END)
.../src/main/java/com/mim736/qvs/QvsApplication.java | 12 ++
.../java/com/mim736/qvs/config/SecurityConfig.java | 75 ++++++++
backend/src/main/java/com/mim736/qvs/domain/Role.java | 7 +
.../main/java/com/mim736/qvs/domain/UserAccount.java | 107 ++++++++
.../com/mim736/qvs/repo/UserAccountRepository.java | 15 ++
.../mim736/qvs/security/JwtAuthenticationFilter.java | 45 ++++++
.../main/java/com/mim736/qvs/security/JwtService.java | 66 ++++++++
.../mim736/qvs/security/QvsUserDetailsService.java | 37 ++++++
.../main/java/com/mim736/qvs/service/AuthService.java | 69 ++++++++
.../main/java/com/mim736/qvs/web/AuthController.java | 35 ++++++
.../java/com/mim736/qvs/web/BusinessException.java | 12 ++
.../com/mim736/qvs/web/GlobalExceptionHandler.java | 44 ++++++
.../java/com/mim736/qvs/web/NotFoundException.java | 12 ++
.../java/com/mim736/qvs/web/dto/AuthResponse.java | 34 ++++++
.../java/com/mim736/qvs/web/dto/LoginRequest.java | 28 ++++
.../java/com/mim736/qvs/web/dto/RegisterRequest.java | 66 ++++++++
backend/src/main/resources/application.yml | 45 ++++++
.../java/com/mim736/qvs/security/JwtServiceTest.java | 25 +++
20 files changed, 935 insertions(+)

```

commit 8e8e77cd9eafb828cc64b18ed3d84cf19242bc2b

Author: **Lyn** Nyandoro <nyandorolyn@gmail.com>
Date: Thu Sep 3 19:51:07 2026 +0200

Initialise the qualification verification repository.

Establish the assignment README and ignore rules so Java, React and Docker artefacts stay out of version control.

Co-authored-by: Cursor <cursoragent@cursor.com>

```
.gitignore | 43 ++++++
README.md | 78 ++++++
2 files changed, 121 insertions(+)
(END)
(base) lynnyandoro@Lyns-Laptop Qualification-Verification-System % git log --oneline --graph --all
* 17ce488 (HEAD -> main, origin/main, origin/HEAD) Removal of user credentials from Home screen
* a29227e Search_by_Name
* 743c8e8 Dark_Light theme togggle bar added
* 838cf28 verification by candidate name
* c10e37b Registrar controls additions
* ee42dbb Team members addition
* 80234be About Us and Contact Us pages added
* 6c396e2 Add campus demo data, mint UI, and distinct credential IDs.
* a90e4cf Promote UI hardening onto main.
|\
| * 34c5ac8 Merge feature/ui-hardening into develop.
| |\
| | * cb34dfb Harden the CRA login flow and pin ajv for webpack.
| | /
| | a64bcd6 Release develop into main for the MIM736 practical.
| |\
| | * ab25c09 Resolve overlapping documentation edits on the conflict log.
| | \
| | | * 1857a3f Clarify audit-trail wording in the conflict-practice note.
| | | * 571ac55 Merge hashing wording into develop.
| | | \
| | | /
| | | 2f153c3 Clarify hashing wording in the conflict-practice note.
| | | /
| | * 07369a1 Merge feature/documentation into develop.
| | \
| | | * 3c712af Document architecture, Git workflow, issues and assessment artefacts.
| | | /
| | * 524daeb Merge feature/cicd-docker into develop.
| | \
| | | * bdbc162 Add GitHub Actions quality gates and Docker Compose delivery.
| | | /
| | * 6bcc28c Merge feature/react-ui into develop.
| | \
| | | * 17eacd1 Add a Create React App interface for register, search, verify and audit.
| | | /
| | * 639d6d5 Merge feature/verification-audit into develop.
| | \
| | | * 785da56 Verify authenticity and persist an append-only audit trail.
| | | /
| | * 9adeb8b Merge feature/qualification-api into develop.
| | \
| | | * d855d10 Register and search qualifications with integrity hashing.
| | | /
| | * 96808b9 Merge feature/auth-and-roles into develop.
| | /
| | * f080609 Add JWT authentication and role-based access control.
| | /
| * 8e8e77c Initialise the qualification verification repository.
```